Topic:- Functions With Argument

Interview Based Questions

1. How do you define a function with a single argument?

Defination:You define it using the def keyword, followed by the function name and the parameter in parentheses. Syntax:

Syntax:

```
def greet(name):
    print(f"Hello, {name}!")
greet("Aarav")
```

2. How do you pass multiple arguments to a function?

Answer: Parameters are separated by commas in the function definition, and arguments are passed in the same order. Syntax:

Syntax:

```
def add_numbers(a, b):
    return a + b
result = add_numbers(5, 10) # 5 goes to a, 10 goes to b
```

3. What are Default Arguments and how are they defined?

Answer: Default arguments are used if no value is provided during the function call. They must follow non-default arguments. Syntax:

Syntax:

```
def describe_city(city, country="India"):
    print(f"{city} is in {country}")
```

```
describe_city("Ahmedabad")        # Uses default "India"
describe_city("Tokyo", "Japan")   # Overrides default
```

4. What are Keyword Arguments?

Answer: These allow you to pass arguments by naming them, meaning the order does not matter.

Syntax:

```
def build_profile(first, last):
    print(f"Name: {first} {last}")
build_profile(last="Patel", first="Smit")
```

5. How do you handle an arbitrary number of positional arguments?

Answer: Use *args (the star operator). This collects extra arguments into a tuple. Syntax:

Syntax:

```
def list_items(*items):
    for item in items:
        print(f"Item: {item}")
list_items("Laptop", "Mouse", "Keyboard")
```

6. How do you handle an arbitrary number of keyword arguments?

Answer: Use **kwargs (double star). This collects extra named arguments into a dictionary. Syntax:

Syntax:

```
def show_scores(**scores):
    for subject, score in scores.items():
        print(f"{subject}: {score}")
show_scores(Math=95, Science=88, English=92)
```

7. What is the difference between "Pass by Object Reference" in Python?

Answer: Python passes references to objects. If you pass a mutable object (like a list) and modify it, the original changes. If you pass an immutable object (like a string), the original stays the same. Syntax:

Syntax:

```
def update_list(my_list):
    my_list.append(4)
nums = [1, 2, 3]
update_list(nums)
print(nums) # Output: [1, 2, 3, 4]
```

8. How do you enforce "Keyword-Only" arguments?

Answer: You can use an asterisk * in the parameter list. Everything after the * must be passed as a keyword argument. Syntax:

Syntax:

```
def set_config(*, timeout, retry):
```

```
    print(f"Timeout: {timeout}, Retry: {retry}")

# set_config(30, 3) -> This would cause an Error

set_config(timeout=30, retry=3) # Correct
```

9. Can a function accept another function as an argument?

Answer: Yes, Python functions are "first-class objects," meaning they can be passed around like variables. Syntax:

Syntax:

```
def shout(text):

    return text.upper()

def greet_shouting(func, name):

    message = func(f"Hello {name}")

    print(message)

greet_shouting(shout, "Vivek")
```

10. What are Type Hints for function arguments?

Answer: Introduced in Python 3.5+, type hints specify the expected data type for arguments to improve code clarity and debugging. Syntax:

Syntax:

```
def calculate_area(radius: float) -> float:

    return 3.14 * (radius ** 2)

print(calculate_area(5.0))
```